Wallmart sues Tesla
After seven roof fires, Walmart sues Tesla over "negligent installation and maintenance " of solar panels. **https://dgit.in/sep19-57**

Stolen IPs and China
According to the CNBC CFO survey, 1 in 5 corporations say China has stolen their IP within the last year. **https://dgit.in/sep19-58**

tape. These specific points or codes were known as G-codes, and these were designed to allocate positioning instructions to the machine. However, the problem with these machines was that they were hardwired. This made it impossible to alter the pre-set parameters. With CNC machines, G-codes were still utilised as a means of control, however, they were now designed, controlled and conducted via computer systems. Now G-codes in CNC machines along with logical commands have amalgamated to form a relatively new programming language called parametric programs. Unlike NC machines, the CNC machines that feature this programming language allow the operator of the machine to make real-time adjustments. CNC machines have increased accuracy, more productivity, efficiency and safety when compared to their older NC variants.

Controls of the NC machines were on specific points on punched tapes

DEVELOPMENT OF CNC MACHINES

After the MIT project, Richard Kegg curated the Cincinnati Milacron Hydrotel in 1952, in collaboration with MIT, which can be considered as the first CNC milling machine. This machine was patented by him under the name 'Motor Controlled Apparatus for Positioning Machine Tool' in the year 1958.

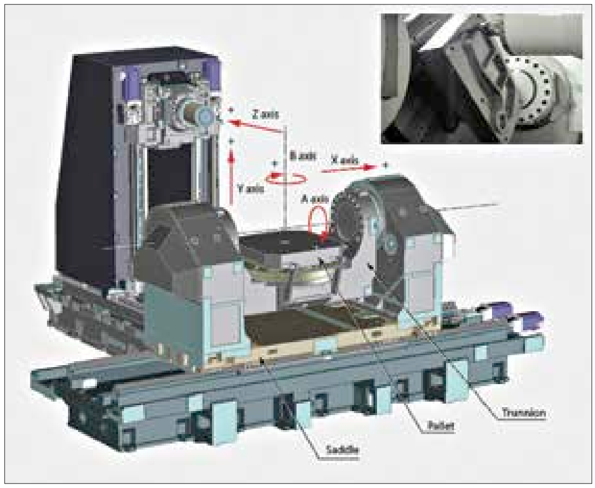
As numerical control technology traversed into the 1960s and 70s, the beginnings of the familiar form of a CNC machine that most would identify today started taking shape. Digital technology began entering the scene, and automation in production processes became more seamless than it had ever been in the past. Today, it is not rare to find CNC machines in a majority of industries, especially with the advancement of computer technology. It has also become easier for people to create their own personal versions of CNC machines.

Despite the advent of CNC machines more than a decade ago, the proliferation of these machines only began in the year 1967. From then onwards, it was commonplace to see the implementation of Computer-Aided Design and Computer-Aided Machining. This also led to the further development of CNC machines since they were in demand owing to the long strides industrialisation was taking. In the year 1976, the very first 3D Computer-Aided design/Computer-Aided Machining systems started becoming available. By 1989, you could clearly comprehend that these machines had become the industry standard.

HOW A CNC MACHINE WORKS NOW

The term machining, in general, means a way of transforming a stock piece of material such as a block of plastic into a finished product, which is typically a prototype part, by using a controlled material removal process. Similarly, CNC relies on Computer-Aided Manufacturing or Computer-Aided Design and interprets the design as instructions for cutting up prototype parts. While machining processes often require using multiple tools to make desired cuts, CNC machines generally combine tools into common units from which the machine is able to draw out the design. Basic CNC machines are able to move about in one or two axes while advanced machines can move laterally in the x and y-axis and longitudinally in the z-axis as well. They often even move rotationally about one or more axes.

Multi-axis CNC machines are actually capable of automatically flipping over parts, which allows the removal of material that was previously 'underneath'. This eliminates the need for manual flipping, hence allows leaving the machines unmanned, as mentioned earlier. These machines allow for the cutting of all sides without any manual intervention at all which is what makes them so desirable and commonplace in industries.



Multi-axis CNC machine

STAPLES OF CNC TECHNOLOGY

While the history of CNC machines has been quite dynamic, undergoing multiple changes, there are a few cornerstones that remained unchanged. All automated motion control machines, right from the barebones concepts back in the day to the advanced machines today, require three primary components. These components include a command function, a drive or motion system and a feedback system. Even when it comes to CNC milling, there has not been too much change from the original model that was created in MIT in 1952 by Richard Keggs. These machines typically consist of a table that moves in the X and Y axes while a tool spindle moves in the Z-axis. While we have multi-axis machines today which allow flipping over of the material, as iterated above, all CNC machines still make use of these fundamentals.